Exercises in *Fourier Analysis on Number Fields*

Solutions to the exercises of Ramakrishnan–Valenza

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Exercise 7.1. Let K be a local field and let ψ be a nontrivial unitary character on K; that is, a continuous, nontrivial homomorphism from K as an additive group into S^1, the circle group. For each $a \in K$, define $\psi_a \colon K \to S^1$ by

$$\psi_a(x) = \psi(ax).$$

(a) Show that ψ_a is again a unitary character of K, and that ψ_a is trivial if and only if $a = 0$.

(b) Show that $a \mapsto \psi_a$ defines a continuous, injective homomorphism α_ψ of K into its Pontryagin dual $\widehat{K}$.

(c) Show that the image of α_ψ is dense in $\widehat{K}$. [*Hint*: Show that $\psi_a(b) = 1$ for all $a \in K$ if and only if $b = 0$.]

(d) Show that α_ψ is bicontinuous.

(e) Show that $\alpha_\psi(K)$ is a complete, hence closed, subgroup of $\widehat{K}$. Conclude that α_ψ is an isomorphism of topological groups.

Solution. (a) Multiplication by a, $x \mapsto ax$, is a continuous endomorphism of the additive group K. Hence ψ_a, its composite with ψ, is a continuous homomorphism $K \to S^1$, i.e. a unitary character of K.

If $a = 0$, then $\psi_a(x) = \psi(0) = 1$ for all $x \in K$, so ψ_a is trivial. Conversely, let $a \neq 0$. Since ψ is nontrivial, there is some $y \in K$ with $\psi(y) \neq 1$, and then $\psi_a(a^{-1}y) = \psi(y) \neq 1$. So ψ_a is nontrivial.

(b) Recall that $\widehat{K}$ has the compact-open topology. A basis of neighborhoods of a character ψ_0 is given by the sets

$$W(\psi_0, C, \epsilon) = \{\psi \in \widehat{K} : |\psi(x) - \psi_0(x)| < \epsilon, \forall x \in C\}.$$

where $C \subset K$ is a compact set and $\epsilon > 0$.

First, reduce to continuity at 0. Note that $\psi_{a+b}(x) = \psi_a(x) \cdot \psi_b(x)$. Then we have that

$$|\psi_a(x) - \psi_b(x)| = |\psi_b| \cdot |\psi_{a-b}(x) - 1| = |\psi_{a-b}(x) - 1|.$$

So it suffices to show that for each compact C and $\epsilon > 0$, there is a $\delta > 0$, such that $\forall |a| < \delta$, $|\psi_a(x) - 1| < \epsilon$.

Given an $\epsilon > 0$. Since ψ is continuous at 0 and $\psi(0) = 1$, then there is an $r > 0$, such that $\forall |x| < r$, $|\psi(x) - 1| < \epsilon$. Since C is compact, then $M = \max_{x \in C}\{|x|\} < \infty$. Choose $\delta = r/M$, then $\forall |a| < \delta$ and $\forall x \in C$, $|ax| < r$. It follows that

$$|\psi_a(x) - 1| = |\psi(ax) - 1| < \epsilon.$$

Injectivity is followed from (a).

(c) Note that

$$\forall a \in K, \psi_a(b) = 1, \Leftrightarrow \forall a \in K, \psi(ab) = 1, \Leftrightarrow b = 0.$$

Recall that we have the fact of Pontryagin dual: let G be a locally compact abelian group and H be a proper closed subgroup of G. Then there is a nontrivial character χ of G, such that $\chi|_H = 1$.

Let H be the closure of the image of α_ψ, i.e. $H = \overline{\alpha_\psi(K)}$. Assume that $H \neq \widehat{K}$. Then we have a character $\chi \in \widehat{\widehat{K}}$ such that χ is trivial on H. And by Pontryagin duality $K \equiv \widehat{\widehat{K}}$, every character of $\widehat{K}$ is evaluation at some point $b \in K$, i.e. $\psi \mapsto \psi(b)$. Then we have that $\chi = (\psi \mapsto \psi(b_0))$ for some $b_0 \in K$. Since χ is trivial on H then $\forall a \in K, \psi_a(b_0) = 1$. It follows that $b_0 = 0$, which contradicts to χ is a nontrivial character of $\widehat{K}$.

□

Exercise 7.2. Let K be a non-Archimedean local field with ring of integers $\mathfrak{o}_K$ and maximal ideal P. Fix a nontrivial unitary character ψ of K. Let P^m be the conductor of ψ; that is, the largest fractional ideal P^m on which ψ is trivial. Let α_ψ, as defined in Exercise 7.1, be the isomorphism between K and its Pontryagin dual.

(a) Show that via α_ψ, P^m identifies with the Pontryagin dual of $\widehat{\mathfrak{o}}_K$. (This is also the inverse different of K, which soon makes its appearance.)

(b) Let $n \geq 1$ and $x \in K$. Show that $xP^n \subseteq \ker(\psi)$ if and only if $x \in P^{m-n}$.

Bibliography

[RV99] D. Ramakrishnan and R. J. Valenza, *Fourier Analysis on Number Fields*, Graduate Texts in Mathematics **186**, Springer-Verlag, New York, 1999.